

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

KRISHA MEENATCHI M (192511338)

Exp-09: Index of First Occurrence in a String

Aim:

To find the index of the first occurrence of a given substring in a string using a brute-force string matching technique.

Algorithm:

1. Read the main string.
2. Read the substring to be searched.
3. Start comparing the substring from every possible position in the main string.
4. Compare each character of the substring with the corresponding character of the main string.
5. If all characters match, return the starting index.
6. If no match is found, display -1.
7. Display the time and space complexity.

Source Code:

```
text = input("Enter the main string: ")
```

```
pattern = input("Enter the substring: ")
```

```
n = len(text)
```

```
m = len(pattern)
```

```
index = -1
```

```
for i in range(n - m + 1):
```

```
match = True
```

```
for j in range(m):
```

```
    if text[i + j] != pattern[j]:
```

```
        match = False
```

```
        break
```

```
if match:
```

```
    index = i
```

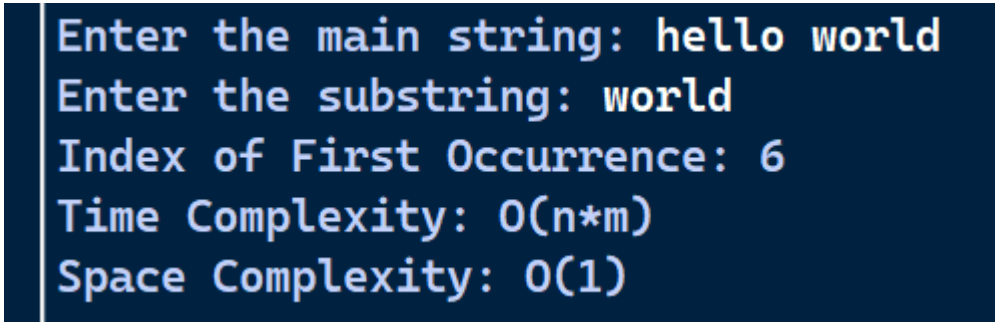
```
    break
```

```
print("Index of First Occurrence:", index)
```

```
print("Time Complexity: O(n*m)")
```

```
print("Space Complexity: O(1)")
```

Output:

A screenshot of a terminal window with a dark blue background and light blue text. The text shows the program's execution: 'Enter the main string: hello world', 'Enter the substring: world', 'Index of First Occurrence: 6', 'Time Complexity: O(n*m)', and 'Space Complexity: O(1)'.

```
Enter the main string: hello world
Enter the substring: world
Index of First Occurrence: 6
Time Complexity: O(n*m)
Space Complexity: O(1)
```

Result:

The index of the first occurrence of the given substring was successfully found using brute-force string matching.